KIET ANH TRAN

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OBJECTIVE

I am seeking an opportunity in robotics or mechanical engineering with a focus on mechanical design, mathematical modeling, control, and autonomous systems. I aim to leverage my background in robotics, control theory, and mechanical design to develop innovative solutions for complex robotic systems and advance research in automation and intelligent machines.

SKILLS

Software: python, pytorch, MATLAB, Labview, SOLIDWORKS, CATIA v5, ROS2, Eagle.
Hardware & Prototyping: 3D printing, CNC machining, Arduino, Raspberry Pi, embedded system design.
Mechanical & Robotics Engineering:

- Mechanical design & analysis: kinematics, dynamics modeling, mechanical design, FEA analysis.
- Robotics & simulation: forward/inverse kinematics, trajectory planning, state estimation, PID, MPC, adaptive control, reinforcement learning.

EXPERIENCES

Research Thesis – Vine Robot Control | Johns Hopkins University, Advisor: Dr. Jin Seob Kim | June 2025 – Current

- Modeled vine-like continuum robots using Cosserat rod theory and biologically inspired growth dynamics.
- Developed simulations to optimize robot–wall interactions and predict the robot's deformation during contact.
- Applied a Model Predictive Path Integral (MPPI) controller to enable real-time control.

Research Intern | **Vietnam Academy of Science and Technology** | July 2023 – July 2024

- Developed a vision-based traffic sign detection and tracking pipeline, training a deep learning model and integrating DeepSORT for persistent multi-object tracking across video streams.
- Implemented an acoustic perception module using a CNN-based vehicle sound classifier, enabling environment awareness under low-visibility conditions where camera sensing degrades.
- Integrated perception, planning, and control components into a unified ROS2-based autonomous driving system.

Research Assistant | Applied Optics Lab | January 2022 – May 2023

- Designed and fabricated lab research equipment using SOLIDWORKS, CAM, and 3D printing.
- Developed a waterproof enclosure for an immersed biosensor and conducted water-resistivity tests.
- Created a modular photonic testbed with custom adapters for sensor calibration and testing.
- Performed calibration tests to validate sensor compatibility and performance across different applications.

PROJECTS

Imitation Learning based controller for flexible guidewire navigation | October 2025 – December 2025 | 1st Prize

- Developed an imitation learning–based control framework for deformable catheter navigation, enabling autonomous steering without relying on explicit kinematic or dynamic models.
- Designed and integrated a physical experimental platform including a soft catheter, linear actuation stage, and syringe-driven pressure system for real-world validation.
- Validated the controller in hardware experiments, demonstrating stable trajectory tracking in confined environments.

Solar Tracking Panel Project | January 2023 - May 2023

- Researched solar tracking panel as a possible solution for improving efficiency of solar panels.
- Conducted market analysis to generate engineering requirements for final products.
- Sketched and simulated stress deformation of final design structure using SOLIDWORKS.
- Built and tested moving and charging function of final solar panel prototype.

EDUCATION

JOHNS HOPKINS UNIVERSITY | Baltimore, MD
Master of Science in Mechanical Engineering | Expected 05/2026 | GPA: 3.65/4.00

PURDUE UNIVERSITY | West Lafayette, IN
Bachelor of Science in Mechanical Engineering | 05/2023 | GPA: 3.58/4.00
Minor: Mathematics